

GaugePredict: An open-source python package for basin-scale gauge forecasting using a hybrid neural network.

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Summary

Operational decisions and scientific analyses often rely on forecasts derived from distributed monitoring networks at days-to-weeks lead times. Although basin-scale forecasting is increasingly feasible, routine use is still constrained by the practical work of assembling reproducible multi-site workflows: retrieving records from many stations, reconciling mismatched time coverage, handling missing data, and producing consistent training and evaluation outputs across multiple forecast horizons. This paper presents GaugePredict, a Python package that standardizes these steps into an end-to-end workflow for multi-site time-series forecasting that can be applied across basins and extended to any environmental variable represented as a site-based time series.

GaugePredict is an open-source Python package that provides an end-to-end workflow for basin-scale time-series forecasting from distributed monitoring networks. Although the primary examples focus on daily river discharge and stage, the software is designed to be parameter-agnostic: any variable available as a multi-site time series (for example, water quality constituents, temperature, or other environmental indicators) can be used as predictors and targets when supplied through NWIS or user-provided files. GaugePredict implements a hybrid convolutional neural network–long short-term memory (CNN–LSTM) model, an architecture widely applied in time-series prediction problems (Lu et al., 2020). A one-dimensional convolutional neural network (Conv1D) extracts local temporal patterns from multivariate inputs and compresses them into feature maps (LeCun and Bengio, 2002), and a long short-term memory (LSTM) network then models longer-range dependencies through gated memory (Hochreiter and Schmidhuber, 1997). This combination is well suited to basin-scale applications because it supports efficient feature extraction from many predictor sites while representing temporal lags and persistence that influence downstream conditions (Lu et al., 2020).

GaugePredict retrieves and preprocesses multi-site daily time series from the U.S. Geological Survey National Water Information System (USGS-NWIS) (U.S. Geological Survey, 2026), aligns predictors and targets (which can be USGS records or user-imported CSV files) in time, constructs supervised learning sequences, trains models in PyTorch (Paszke et al., 2019), and produces standardized evaluation outputs and figures. For interpretability and model simplification, GaugePredict

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uses SHapley Additive exPlanations (SHAP), a game-theoretic attribution method that assigns each predictor site a contribution to a model prediction (Lundberg and Lee, 2017). These attributions are summarized to rank predictors and optionally define reduced-input models, which can lower computational cost while preserving forecast behavior and support workflows where lightweight short-lead forecasts are updated frequently on standard local machines.

Statement of Need

Basin-scale, data-driven gauge forecasting is increasingly feasible, but practical adoption is often limited by workflow friction rather than model availability. Users typically need to (i) gather multi-gauge records from a data provider, (ii) reconcile inconsistent time coverage and missing values, (iii) build consistent training and evaluation setups across multiple lead times, and (iv) understand which gauges provide predictive value at different horizons. These steps are frequently assembled ad hoc, making it difficult to reproduce results, compare horizons fairly, or transfer a workflow to another basin or target gauge.

GaugePredict addresses these barriers by providing a transparent, end-to-end pipeline for daily gauge forecasting that is designed for rapid application and adaptation. The package emphasizes reproducibility through configuration-driven runs and saved artifacts, and interpretability through SHAP-based gauge ranking that can be used to reduce the predictor network while retaining comparable forecast behavior across lead times.

Software Description

Functionality

GaugePredict is organized around common user tasks in basin-scale forecasting:

- **Data retrieval and preprocessing:** retrieve daily values from USGS-NWIS using the Python `dataRetrieval` ecosystem (U.S. Geological Survey, 2026), assemble basin-wide predictor time series, align predictors and a target gauge record on a shared daily index, apply consistent train/test splits, and handle missing data in a reproducible way. Target site data can be USGS records or user-imported files.
- **Model training and evaluation:** train CNN-LSTM models for user-selected lead times using a standardized training interface, report comparable metrics, and export model artifacts (trained weights and preprocessing objects) for reuse using PyTorch (Paszke et al., 2019).
- **Explainability and predictor reduction:** compute SHAP values to attribute predictions to individual predictor gauges, summarize gauge importance, and optionally train reduced-input models using top-ranked gauges (Lundberg and Lee, 2017).
- **Visualization and diagnostics:** generate training diagnostics, observed versus predicted time series, and SHAP summary and geospatial context plots.

Design and architecture

GaugePredict prepares aligned predictor and target arrays and constructs sliding-window sequences for supervised learning. A training wrapper manages lead-time-specific training, metric logging, and artifact export, supporting consistent comparisons across horizons. Plotting utilities standardize performance figures and SHAP-based summaries so users can rapidly assess skill, errors, and predictor influence.

The software is organized into modules to support use and extension:

- **downloader**: retrieval of NWIS daily values and construction of basin-scale predictor catalogs.
- **routines**: shared utilities for preprocessing, alignment, splitting, and input/output.
- **predict**: high-level functions for training and inference at specified horizons.
- **plotting**: standardized evaluation and interpretability figures (performance, SHAP summaries, maps).

Example Usage

A minimal 7-day forecast workflow:

```
from GaugePredict.predict import run_horizon

results = run_horizon(
    forecast_horizon=7,
    data_files="path/to/predictor/files",
    use_csv_target=False,
    target_site="07374000",
    target_parameter_code="00060",
    start_date="2020-01-01",
    end_date="2023-12-31",
    tz="UTC",
    hp={"epochs": 50, "batch_size": 32, "sequence_length": 30},
    device="cuda" # or "cpu")

print(results["metrics"])
```

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